CardioTwin AI — System Architecture & Data Flow Diagram

Digital Twin Challenge 2026 | Happiest Health Reimagining Healthcare Summit, Bengaluru

Team Phoenix: Vansh Singla (Lead), Dev Garg, Prince Pal | **Maharaja Agrasen Institute of Technology (MAIT), Delhi**

STREAM 1: STATIC EHR	STREAM 2: DYNAMIC WEARABLES	DATA INGESTION & PIPELINE	AI / ML MODELS & PREDICTION	CLINICIAN DIGITAL TWIN UI
• Synthea Synthetic EHR Engine • Patient Demographics (Age, Sex) • Chronic Comorbidities (HTN, T2D) • Lab Observations & BMI • Medication Status	• PhysioNet BIDMC PPG (125 Hz) • PhysioNet MIT-BIH ECG (360 Hz) • Continuous Plethysmograms • Real-Time BLE IoT State Machine • Wearable Photoplethysmography	• SciPy 3rd-order Butterworth Filter (0.5–5.0 Hz bandpass) • Real-time Peak Detection • Signal Quality Index (SQI) • Negation-aware NLP Parser • Strict Provenance Isolation	• **Multimodal Late-Fusion Engine** • **MIT-BIH Arrhythmia Model:** Supervised Random Forest (AAMI) • Arterial Pressure & BP Forecast • Explainable AI (XAI) Attribution • SQLite Longitudinal Memory	• 3D Glowing Digital Twin Avatar • 4 Live Telemetry Hotspots • Dynamic Symptoms & Complaints • Checkup Timeline & Consults • Stamped ReportLab PDF Export • Contextual AI Chatbot

DATA PRIVACY, SECURITY & DPDP ACT COMPLIANCE LAYER:

Provenance Classification	Data Security & Isolation	Algorithmic Transparency
• **OPEN_RESEARCH_DATA:** PhysioNet BIDMC & MIT-BIH open databases. • **SYNTHETIC_EHR:** Synthea model profiles; visibly flagged as synthetic. • **AUTHORIZED_PATIENT_DATA:** User entries stored locally in SQLite.	• Zero data transmission to third parties. • Local execution on patient premises. • Complies with Digital Personal Data Protection (DPDP) Act 2023 & HIPAA. • Zero silent mock fallbacks active in production.	• Explainable AI (XAI) feature contribution breakdown. • AAMI standard beat classification metrics. • Participant-level train/test separation (zero data leakage). • Prominent clinical supervision prototype notice.